INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

AUTOMATED MANUFACTURING MACHINE, ABRASIVE WATERJET MACHINING CENTER, COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numerical control (CNC) abrasive waterjet machines intended to accurately cut shapes from stock material, primarily metal. Abrasive waterjet machining is an industrial cutting process that pumps a mixture of high-speed, high-pressure water and an abrasive media (garnet) through a work piece, imparting little to no additional heat on the item. The cutting jet is approximately 90 percent water and 10 percent garnet by mass flow, has a diameter of approximately 0.03 inches, and ranges in pressure between 40,000 and 90,000 pounds per square inch (psi). CNC refers to a computer-controlled system able to programmatically command and precisely control all machine movements and functions.

The term “waterjet” as used in the remainder of this document refers to the system in its entirety including the cutting table, pump, controller, and any defined peripheral equipment. The waterjets identified herein shall be manufactured within the United States of America and shipped as a complete unit. The waterjets shall be capable of cutting mild steel at least 4 inches thick as well as aluminum, Plexiglas, titanium, copper, granite, wood, glass, and hardened tool steels.

CNC machining is an automated machine process used to manufacture components by selectively and precisely removing material from a workpiece. Parts manufactured via CNC machining are most practical via use of Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM) software. CAD enables development and design of a part. CAM programs are used to generate manufacturing tool paths and to automate the manufacturing processes; CAM converts the CAD model into an executable code or program for machine operations.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28, Wright-Patterson AFB, OH 45433 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

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2. CLASSIFICATION. The waterjets shall be of the size and type specified below. The type to be furnished shall be as specified in 3.2.1 and the size shall be as specified in 3.2.2. Note that Sizes I and II are the only two available, as Type I machines. While Sizes III and IV are only available as Type II machines.

Type I – cantilever-style cutting support.
Size I – smallest cutting envelope (4'7" x 4'7") for typical half sheet.
Size II – medium cutting envelope (8'4" x 4'7") for typical full sheet.
Type II – bridge-style cutting support.
Size III – large cutting envelope (10'6" x 5'2").
Size IV – maximum cutting envelope (13'4" x 6'8").

3. SALIENT CHARACTERISTICS.
3.1 General requirements. The waterjet shall be new (not a rebuilt or remanufactured) and one of the manufacturer's standard current production models (not a prototype) capable of operations in accordance with the requirements herein. The waterjet shall be constructed from materials with sufficient mass and strength and configured in a shape and size that supports the workpiece and machine components while minimizing element deflections and initiated and/or transmitted vibration during operation. Designs with a separate catcher tank (independent of cutting unit) shall not be allowed. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Access doors or removable panels shall be provided for servicing all components located within the waterjet.
The waterjet shall be delivered as a complete system so that when connected to the utilities identified herein, it can be used for any function for which it is designed and constructed. As such, delivery of the waterjet must include all tooling, accessories, internal and external hardware, software, and components required for installation and operation.

3.2 Components: The waterjet shall consist of, but not be limited to, the principal components, attachments, and accessories necessary to meet the operational and performance requirements as defined below:

a. Cutting Assembly Support (3.2.1).
b. Machine Table (3.2.2).
c. Controller (3.2.3).
d. Operating/Cutting Software (3.2.4).
e. Cutting & Movement Operations (3.2.5).
f. Pump with Pump Rebuild Kits (3.2.6).
g. Garnet Storage Hopper (3.2.7).
h. Lathe/Rotational Head (3.2.8).
i. Cutting Head (3.2.9).
j. Chiller System (3.2.10).
k. Water Recycling System with Rebuild Kit (3.2.11).
l. Garnet and Solid Removal System (3.2.12).
m. Remote Keypad (3.2.13).
n. Water Level Control (3.2.14).
o. Nozzle Assembly with Replacement Nozzle Assemblies (3.2.15).

3.2.1 Cutting assembly Support. The cutting head/nozzle assembly shall be mounted above the machine table in a manner maintains the tolerances and repeatability established in section 3.8 The support structure for the cutting head/nozzle assembly shall be rated a minimum of two times the static and operational loads based on the yield strength of all load-bearing components. The support components shall support precision movement of the cutting head/nozzle assembly

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and rigidly hold the cutting head/nozzle assembly in place, preventing unintended movement due to normal machine operations or bumping of the equipment by operating personnel. The cutting assembly support shall have 360 degree wrap-around bellows to prevent garnet and water entry into the drive system. Movement of the cutting support assembly shall be controlled via closed-loop, digital drives with brushless servo motors that require no periodic tuning.

3.2.1.1 Type I head assembly. For Type I machines, the cutting head/nozzle assembly shall be supported in a cantilever configuration where the assembly is attached to only one side of the machine table.

3.2.1.2 Type II head assembly. For Type II machines, the cutting head/nozzle assembly shall be supported in a bridge configuration where the assembly is attached to two, opposite sides of the machine table.

3.2.2 Machine table. The machine footprint, working envelope, maximum load and cutting travel shall be per Table I. The bed shall have four inch x 14 gauge (minimum thickness) galvanized steel support slats. The bed shall have means for leveling the machine on a flat mounting surface. The system must be of unitized design to eliminate misalignment due to an operator bumping either the cutting head or table during material loading. The base shall have access doors or removable plates as necessary for servicing components housed within. The contractor shall provide the necessary leveling hardware to insure compliance with machine specifications. All structural components for the table shall be made of material or coated/painted to prevent corrosion.

TABLE I: Machine Size and Requirements.

	Size I	**Size II**	**Size III**	**Size IIII**
TYPE				
Machine Type (See 3.2.1)	Type I	Type I	Type II	Type II
TRAVELS				
Min. X-Y Travel	4’7” x 4’7”	8’4” x 4’7”	10’6” x 5’2”	13’4” x 6’8”
TABLE				
Min. Table Size (W x L)	6’8” x 5’5”	10’6” x 5’5”	12’ x 6’	15’ x 7’
Max Waterjet Footprint (includes control station; excludes pump, garnet storage, and other accessories)	11’ x 8’	13’ x 8’	19’ x 10’	23’ x 13’
Maximum Operating Weight	15,000 lbs.	25,000 lbs.	25,000 lbs.	35,000 lbs.
Maximum Supported Material (workpiece) Load	400 lbs/sq. ft.	400 lbs/sq. ft.	400 lbs/sq. ft.	400 lbs/sq. ft
NOZZLE				
Min. Z-axis travel (see 3.2.9)	7.0 inches	7.0 inches	6.75 inches	6.75 inches
SPEED				
Minimum Traverse Speed	180 inches per minute	180 inches per minute	180 inches per minute	180 inches per minute

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3.2.3 Controller. The waterjet shall be provided with a fully automatic, solid-state, integrated circuit, 64-bit, multiple microprocessor PC-based controller that runs on Windows® 10 operating system. The controller shall include a minimum 17-inch, LCD, LED, OLED, or similar flat panel display, standard mouse and keyboard inputs, and a minimum of two additional USB ports not utilized by the mouse or keyboard. The controller and screen shall be located in a water-resistant housing that is attached to the waterjet frame at a height suitable for operating while standing. The controller shall be capable of saving, storing, and recalling all associated operating files (see 3.2.4) to/from an internal hard drive. The controller shall be capable or reading/importing all associated operating files from an external disk drive (CD/DVD) connected via USB. The controller shall be capable of full operations without an internet or network connection.

3.2.4 Software. The waterjet shall ship from the vendor with all software required for operations installed on the controller as defined in 3.2.4.1 and 3.2.4.2. The software shall provide full operational control of all waterjet machining functions. The software shall be fully integrated into the controller (see 3.2.3). The software shall be capable of update/upgrade without the controller requiring an internet or network connection. The vendor shall provide three (3) additional sets (with operating licenses) of the associated software.

3.2.4.1 2-D Functionality. The software for Type I and Type II machines shall provide full cutting/jetting functionality in two dimensions (2-D, flat) across the machine table. These functions shall include: automatic and manual lead in/out, parametric shape generator, automatic collision prediction and correction, support tool path fonts, Windows® true type fonts, automatic raster-to-vector conversion, and corner and piercing optimizations. The software shall provide the ability to resize, rotate, mirror, align geometric features (individually or as part of a group), and alignment with other features. The software shall automatically select default cut/jet parameters (i.e. – nozzle speed, jet pressure, tool path, pierce locations, over/undercuts, etc) based on input geometry, material type, thickness, and desired finish. The software shall allow for programming/modification of all cut/jet parameters. The software shall calculate cut/movement paths prior to machining operations and provide machine time estimates correct with 1 percent of actual operations, and shall allow for customizable pause and alert based on machine parameters. The software shall be capable of importing models and geometries in the following file formats/extensions: DWG (.dwg), DXF (.dxf,) Initial Graphics Exchange Format (.igs, .iges), g-code (.n), Numerical Control Machine Instructions (.nc), Computer Numeric Control File (.ncc), Portable Document Format (.pdf), and AutoDesk (.ps).

3.2.4.2 3-D Functionality. The software for Type II machines shall provide additional functionality for three dimensional (3-D) cutting/jetting up to 60 degrees in all directions from normal to the machine table. The software shall be capable of importing 3D CAD models in the same formats listed in 3.2.4.1 as well as native CATIA (.CATPart, .CATProduct, .exp, .session, .model), CREO (.asm, .prt,), SolidWorks (.sldprt, .sldasm) and neutral CAD step (.step, .stp), parasolid (.x_b, .x_t, .xmt_bin, .xmt_txt) and Stereolithographic (.stl). The software shall also support identification and export of 2D features from 3D models.

3.2.5 Cutting and Movement Operations.

3.2.5.1 Automated material standoff. For Type I and Type II waterjets, the system shall include integrated capabilities (software and hardware) to automatically adjust the standoff distance to compensate for changes in the material height once the standoff distance has been set at the start of a machining cycle. The system shall also detect potential collisions between the waterjet hardware (i.e., nozzle assembly, etc.) and material and pause the machine to prevent collision.

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3.2.5.2 Automated part/tool alignment. For Type II waterjets, the system shall include integrated capabilities (software and hardware) to automatically identify geometric features (i.e., holes, edges, center points, corners, etc.) on pre-machined material prior to waterjet operations. The system shall also automatically rotate/orient the tool path (cutting/machining program) to precisely align with pre-existing features on the material. The system shall also have the capability to retain images of the feature locations for future reference. The system shall allow for operator/programmer intervention and modification at any point during feature location or cutting program alignment process. Further, the system shall include the capability to save and export identified geometry in neutral CAD format (i.e., reverse engineering).

3.2.6 Pump with pump rebuild kits.

3.2.6.1 Pump requirements. The pump shall be capable of obtaining and maintaining the designed operating pressure, 60,000 pounds per square inch (psi), without the use of hydraulic assistance or pressure attenuators. The design shall support a minimum 500 operational hours between pump rebuilds with a minimum design life of 1,000 operational hours between rebuilds under normal operating conditions. Pumps shall meet the requirements provided in Table II for all waterjet Types. The pump shall include an integrated variable frequency drive that is controlled by the waterjet controller that allows for reduction or modification of pump operating parameters below the maximum threshold to meet cutting/jetting requirements while maximizing efficiency. The pump shall include a 110V, single-phase electrical outlet to power controller and table operations.

TABLE II: Pump Flow Rate Requirements.

Operating Power (Pump Rating)	50 horsepower
Full Load Power Draw	65A at 480VAC (60Hz) or 79A at 380VAC (50Hz)
Output Pressure	60,000 psi
Orifice Opening	0.016 inch
Flow Rate (in gallons per minute) with Orifice Opening and at Output Pressure	1.2 gpm
Pump dimensions in inches (length x width x height)	66 x 36 x 38.5

3.2.6.2 Pump rebuild kits. The waterjet shall be delivered with all special tooling required to disassemble, service, and rebuild the pump for continued use due to wear/tear from normal operations. The waterjet shall be delivered with two (2) pump rebuild kids that shall include all filters, lubricants, seals, single-use connectors, and all other items to reset the service life of the pump to zero (0) hours when rebuilt.

3.2.7 Garnet storage hopper. A garnet storage hopper capable of holding a minimum of 600 pounds of garnet/abrasive shall be included and incorporated into the design. The tank shall be capable of being pressurized up to 125 psi. The garnet storage hopper tank shall include the appropriate air valves and pressure regulators to provide pressure sufficiently and consistently needed to move garnet from the tank to the cutting head/nozzle assembly. The garnet storage

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hopper operations shall be fully compatible and integrated with the waterjet controller and software.

3.2.8 Lathe/rotational head. The waterjet shall include a rotary axis capable of holding/rotating square and round materials for the purpose of cutting tubes, bar stock, and pipe. The lathe/rotational head shall be waterproof, fully submersible, and fully integrated into the waterjet controls and software. The lathe/rotational head shall be removable. Other functionality of the waterjet shall not be altered or prevented by removal of the lathe/rotational head.

3.2.9 Cutting Head.

3.2.9.1 Type I cutting head. For Type I systems, the waterjet cutting head shall have a rotating/angling capability for the purpose of offsetting lead-in taper at the onset of the cut. Taper offset functionality shall be integrated into the system (controller and software). The cutting head tilt from perpendicular to the machine table surface shall be a minimum of ±9 degrees. The cutting head shall also include fully integrated, motorized motion in the Z-axis (vertical) for a total movement no less than 7 inches.

3.2.9.2 Type II cutting head. For Type II systems, the waterjet cutting head shall have a fully integrated (controller and software controlled), multi-axis cutting head. The multi-axis cutting head shall include the capability for taper offset and shall be able to cut a minimum of 60 degrees off perpendicular in all directions. The positional accuracy of the cut angle shall be ±0.09 degrees. The cutting head shall also include fully integrated, motorized motion in the Z-axis (vertical) for a total movement no less than 6.75 inches.

3.2.10 Chiller system. The system shall be provided with a water chiller system for cooling water prior to entering the high-pressure pump. The water chiller system shall be capable of a minimum cooling rate of 60,000 BTU/hour and maintaining pump inlet water temperature of 65 °F or below. The water chiller system shall utilize refrigerant R-407c. The chiller system shall operate at a rated load of 11 amps on 480VAC, 60Hz, 3-phase power. The chiller shall not exceed the dimension of 48 inches long by 30 inches wide by 66 inches high.

3.2.11 Water recycling system with rebuild kit.

3.2.11.1 Water recycling system. The waterjet shall include a water recycling system that allows for the reuse of water and eliminates the need for constant water supply and drain for operations. The water recycling system shall draw water from the machine table/tank and shall filter and condition the water using the following: mechanical filtration to remove solid particles, nano filtration to remove dissolved solids, and bacteria elimination. The water recycling system shall function as an integrated system with the machine table/tank, pump, and chiller system. The system shall include a settling tank with a clean-out of 1 inch National Pipe Thread (NPT) for service The settling system within the water recycling system shall include a submersible sediment pump with ground fault circuit interrupter (GFCI) that moves sediment laden water to a hopper or catcher tank (also included). The water recycling system shall operate at a rated load of 6 amps on 480VAC, 60Hz, 3-phase power. The water recycling system shall not exceed the dimension of 80 inches long by 36 inches high by 74 inches high.

3.2.11.2 Water recycling rebuild kit. The waterjet shall be delivered with all special tooling required to disassemble, service, and rebuild the water recycling system for continued use due to wear/tear from normal operations. The waterjet shall be delivered with two (2) water recycling filter kids that shall include all cartridges, filters, lubricants, seals, single-use connectors, and all other items to return the water recycling system to optimal performance.

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3.2.12 Garnet and solid removal system. The waterjet shall include a system capable of removing garnet, solids, and other sediment from the bottom of the machine table/tank. The removal system shall utilize programmable controls that allow customization of the garnet evacuation rate, run time, and the ability to reverse flow. The garnet evacuation rate shall be capable of reaching a rate of up to 5.6 pounds per minute. The removal system shall evacuate all garnet, solids, and other sediment into a mobile hopper, which is also included. The removal system shall operate at a rated load of 10 amps on 220/240VAC, 60Hz, single-phase power.

3.2.13 Remote keypad. The waterjet shall include a remote keypad/pendant that allows the operator to precisely locate the nozzle and cutting head observing their exact location in relation to the material and standing away from the control panel. The remote keypad shall not provide a method to begin cutting. Cutting shall be initiated only at the controller.

3.2.14 Water level control. The waterjet shall include methodology to adjust the water level in the machine tank relative to the machine table. There shall be at least 3 inches difference between the minimum water level and the maximum water level. The water level control shall be an independent hardware control (switch, lever, buttons, etc.) and not imbedded in software or programming.

3.2.15 Support and nozzle assembly with replacement nozzle assemblies. The cutting head shall be supplied with a one-piece nozzle body and shall be warranted to last 500 hours (min). The waterjet shall be supplied with five (5) replacement nozzle bodies, all of which are one-piece bodies and warranted to 500 hours.

3.3 Electrical System. All electrical equipment shall comply with NFPA-79. The machine shall be designed to operate from a single 125-135A, 440-480V, 3-phase, 60-Hz (±1 Hz) power source. The system shall have a distribution/transformer panel that receives the facility power and transforms, conditions, and distributes appropriate power to the individual systems/components. The panel shall be certified and stamped by Underwriters Laboratory (UL) and shall have a breaker to stop power to all waterjet systems/components. It is acceptable for the panel to be located separate from the waterjet systems/components.

3.4 Construction. The abrasive waterjet cutting machine shall be constructed of parts which are new, without defects, and free of repairs. The cutting axes and tank shall be one unit to guarantee the cutting accuracy listed herein. The machine structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without distortion. Processes such as welding, plugging, or filling with solder or paste shall not be used for reclaiming any defective part(s). All screws, pins, bolts, and other fasteners shall be installed in a manner as to prevent change of tightness.

3.4.1 Castings/forgings. All castings/forgings shall be free from defects, scale, or mismatching. Processes such as welding, peening, plugging, or filling with solders or paste shall not be used on casting or forgings for reclaiming any parts of the machine.

3.4.2 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.4.3 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be clean and free from sand, dirt, fins, sprues, flux, and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be either rounded or beveled unless sharpness is required to perform a function.

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3.4.4 Gears. All gears and pinions, if applicable, shall be designed and manufactured of a suitable ferrous material with proper width and size to transmit repeated, full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.4.5 Painting. The machine shall be painted in accordance with the manufacturer's commercial practice. Yellow or orange safety color-coding shall be used for designating physical hazards.

3.4.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.4.7 Interchangeability. All replaceable parts shall be manufactured to common, published standards (i.e., ASTM, SAE, NEMA, etc.) with tolerances and clearances such that said parts components can be replaced or adjusted without modification of the equipment.

3.4.8 Maintainability. The machine shall be designed and constructed to permit effective maintenance. The machine shall be equipped with access covers (if applicable) to facilitate inspection, cleaning, and repair of parts.

3.5 Equipment training. A complete overview of the machine controller, graphics display, and machine function and operation shall be provided at the installation location. Training shall be coordinated with the training point of contact (POC) at the installation location no later than seven (7) calendar days after setup/commissioning at the installation location. The training POC will be provided by the government at the time of order. Training shall follow installation of the equipment and last no more than three (3) days for a maximum of five (5) technicians. Training shall provide the necessary skills and knowledge to operate the machine safely, execute Manual Data Input functions, manual operation of machinery, load tooling into tool changer, conduct tooling offsets, and importing manufacturing files. Electronic versions of training materials shall be accessible to training participants for a minimum of three (3) years after completion of the training. Training shall also include how to properly service and maintain the main pump, water recycling system, and nozzle assembly.

3.6 Warranty. A minimum one (1) year warranty shall be provided to protect against defects, breaks, or component failures during normal operations. The warranty shall include the replacement and installation of components/assemblies at the installation location by an OEM-certified technician. The warranty does not cover consumable items such as nozzles (3.2.15) or rebuild kits (3.2.6.2 and 3.2.11.2). For equipment located in austere operating environments, an OEM-certified technician may oversee the repair by a government representative; however, replacement components will still be provided as part of the warranty.

3.7 Support. Technical support for the equipment and software shall be provided for a period of three (3) years after the date of purchase. Technical support shall be accessible, at a minimum, via phone and email.

3.8 Performance. When tested in accordance with section 5.3, the waterjet shall meet the precision or standard performance category requirements shown in Table III.

TALE III: Accuracy and Repeatability.

AXIS POSITIONING	MINIMUM REQUIREMENT
X, Y, & Z Linear axes	±0.001”
AXIS REPEATABILITY	
X, Y, & Z Linear axes	±0.001”
BALL BAR TEST	
12” Ball Bar Circle	±0.0030”

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3.9 Safety and health requirements. Covers, warning signs, guards, and other safety devices shall be provided for all parts of the machine that present safety hazards and shall prevent unintended contact with the guarded component/area. The safety devices shall not interfere with operation of the machine. Safety devices shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine shall comply with US Department of Labor 29 CFR 1910.

3.9.1 High pressure tubing. The tubing that supplies pressurized water from the pump to the cutting head shall be stainless steel and shall have a pressure rating at least 2.5 times the normal operating pressure of the system.

3.9.2 Caution-warning plates. Corrosion resistant “Caution” or “Warning” plates shall be securely attached to the equipment in visible locations, with any safety precautions to be observed by the operator or maintenance personnel permanently marked on the plates.

3.9.3 Audible noise. Audible noise emitted by the equipment under no load shall not exceed 81 dBA at the operator’s workstation or at any other point at a distance of three (3) feet from the equipment as measured on the “A” weighted scale of a standard Type II sound level meter.

3.9.4 Prohibited materials. Asbestos, mercury, lithium or polychlorinated biphenyl (PCB) shall not be used in any part or component of the system. Lithium batteries used in the computer system are exempt from this requirement.

3.9.5 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at the location of installation.

3.9.6 Stop switch. An emergency stop switch/button shall be provided that when pressed will instantly stop all motion of the machine and will cease flow through the nozzle.

3.10 System of units. All measuring and indicating devices such as dial indicators, pressure gages, temperature indicators, depth stops, and other similar devices used on the machine shall be graduated in inch-pound (IPS) units. Dual markings with metric and IPS units is acceptable provided the IPS units are clearly discernable. Marking on all measuring and indicating devices shall be permanently and legibly engraved or etched on a non-glare background. All dials and other type indicating devices shall be securely mounted in place.

3.11 Nameplate. A corrosion-resistant, metal nameplate shall be securely attached to the machine and contain the following information:

a. Nomenclature.
b. Manufacturer’s name.
c. Serial number.
d. Machine model designation.
e. Power Input (volts, total amps, phase, and frequency).
f. Short-circuit/overcurrent rating.
g. Contract number or order number.
h. Applicable National Stock Number (NSN).
i. Date of manufacture.

3.12 Item Unique Identification Marking. An Item Unique Identification (IUID) plate shall be provided in accordance with MIL-STD-130N. The IUID plate shall be permanently affixed to the

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machine and shall include, at a minimum, the information specified in Section 3.11. This data on the IUID shall be Machine Readable Information (MRI) for item identification and automatic data capture. It is permissible to combine the nameplate (3.11) and the IUID provided both Human Readable Information (HRI) and MRI are present.

3.13 Lubrication plate or chart. A lubrication plate or chart shall be permanently and securely attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

3.14 Technical data.

3.14.1 Manuals/drawings. The vendor shall furnish one (1) hard copy of all technical manuals/drawings containing all information necessary for the programming, operation and maintenance (parts lists, electrical schematics, ladder logic and I/O diagrams) of the waterjet. One (1) complete set of manuals/drawings shall be sent to the receiving activity concurrent with machine delivery.

3.14.2 Installation drawings. The vendor shall provide to the receiving facility, engineering installation drawings showing the necessary compressed air, water, and electrical utilities of the waterjet.

3.14.3 Documentation. All documentation, i.e. operation manuals, maintenance manuals, programming manuals, student training materials, drawings, vendor literature, laminated note cards, metal name and service plates, service bulletins, upgrade bulletins, and all associated charts/graphs, shall be written in the English language and grammatically correct.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/vendor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operational use of the item.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance Provisions. The products shall meet the salient characteristics (section 3) of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same machine offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The waterjet shall be inspected for compliance with all requirements specified in this CID.

5.3 Testing. The waterjet shall be tested in accordance with sections 5.3.1 through 5.3.3. Should a malfunction or a failure occur during any test, the problem shall be corrected and the test shall be rerun from the beginning.

5.3.1 Test setup, and conditions. The manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall

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provide test instruments and equipment with calibration traceable to the National Institute of Standards and Technology (NIST) and shall provide certification of calibration. The machine shall be tested in the environmental conditions considered by the manufacturer as standard for machine-performance verification.

5.3.2 Operational test. The waterjet shall be operated in accordance with the manufacturer's standard operational test procedures. The waterjet shall be operated for not less than four hours in which proper operation of all controls and features shall be verified.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with procedures considered by the manufacturer as standard for machine performance verification.

5.4 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the performance requirements shown in Table III. Failure of the machine to meet the performance category requirements shown in Table III shall be cause for rejection.

5.5 Responsibility for Inspection. The vendor shall be responsible for the performance of all inspection requirements (examinations and tests) as specified herein. The vendor may utilize his own or any other facility suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements. The vendor shall provide all tooling, fixturing, measurement indicating devices, material and programs required of the compliance, accuracy alignment and machining acceptance tests.

6. PACKAGING Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES.

7.1 Sources of referenced documents.

7.1.1 The United States Department of Labor. Copies of CFR can be obtained at https://www.dol.gov/.

7.1.2 ASSIST - Defense Logistics Agency (DLA). Copies of Military Standards (MIL-STD) can be obtained at https://assist.dla.mil.

7.1.3 Federal Acquisition Regulation (FAR). Copies of FAR may be obtained at https://www.acquisition.gov/browse/index/far.

7.1.4 National Institute of Standards and Technology (NIST). Copies of NIST requirements can be obtained at https://www.nist.gov.

7.1.5 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/.

7.1.6 National Electrical Manufacturers Association (NEMA). Copies of NEMA standards may be obtained at https://www.nema.org/.

7.1.7 National Fire Protection Association (NFPA). NFPA documents may be obtained at https://www.nfpa.org/.

7.1.8 SAE International (SAE). SAE documents may be obtained at https://.sae.org/.

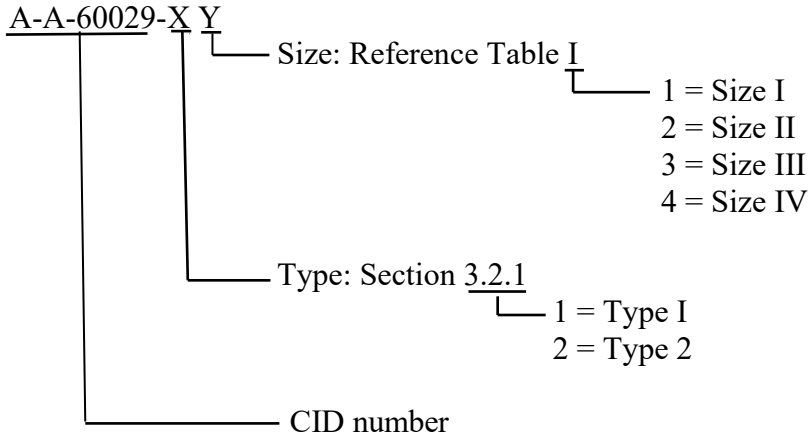
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7.1.9 ASTM International (ASTM). ASTM documents may be obtained at astm.org/.

7.2 Ordering data. Acquisition documents must specify the following:

a. Type of waterjet.
b. Size of waterjet.
c. Point of contact for training coordination at installation location (name, address, email, and telephone numbers).
d. Training POC at Installation Location.

7.2.1 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



7.2.2 Applicable National Stock Numbers (NSNs). The following NSNs are established to meet the requirements of the associated PINs.

PIN	NSN
A-A60029-11	3449-01-702-7792
A-A60029-12	3449-01-702-7793
A-A60029-23	3449-01-702-7795
A-A60029-24	3449-01-702-7794

7.3 Key Words.
Water
Garnet
Cutting

MILITARY INTERESTS:
Custodian:
Air Force – 184

Reviewer:
DLA-GS

Preparing Activity:
Air Force – 184

Agent:
Air Force – 110

(Project 3419-2022-002)

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